# PV101 Day 4 Afternoon Lab

Build a small solar system from five training kits

CTS Cohort 6 - PV101 | Day 4 | Thursday, July 9, 2026

Topic: Small Solar System Build: PV Panel, Charge Controller, 12.8V LiFePO4 Battery, Wiring, and Inverter

Class setup: 6 students; 5 kits. 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.

## Lab mission

- Build one small solar system per kit: 1 PV panel, 1 charge controller, 1 12.8V LiFePO4 battery, wiring, and 1 inverter.
- There are 5 kits and 6 students. One kit is shared by two students. That pair uses Builder/Inspector roles and swaps halfway.
- Success is not speed. Success is a system that can be traced, measured, protected, energized, shut down, and explained.

## Required safety rules

- Polarity is uncertain or readings do not match expectation.
- PV panel is producing voltage while students are still changing wiring.
- Battery fuse blows, breaker trips, arc/spark appears, or equipment smells hot.
- A conductor is warm, damaged, loose, or has exposed copper outside a terminal.
- A student wants to change wiring while anything is energized.
- Inverter wiring path is unclear or routed through the wrong controller terminals.
- The group cannot explain the full energy path clearly.

## Build sequence

1. **0. Assign stations:** 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.
2. **1. Make it safe:** PV panel covered/face-down or unplugged. Battery fuse removed/open. Inverter off. Safety glasses on.
3. **2. Lay out the system:** Place PV panel, charge controller, LiFePO4 battery, protection, inverter, and wires so every connection can be seen and traced.
4. **3. Identify terminals:** Find and label controller BAT +/-, controller PV +/-, battery +/-, inverter +/-. Nobody trusts color alone, because colors have betrayed humanity repeatedly.
5. **4. Prep wires:** Cut/route wires with strain relief and service slack. Strip cleanly. Crimp terminals. Tug test every crimp.
6. **5. Wire battery to controller:** Connect battery side to controller BAT terminals with correct polarity. Keep fuse removed/open until instructor approval. Battery connects before PV for most controllers.

7. **6. Meter battery side:** Measure battery voltage directly, then at controller battery terminals. Expected battery is about 12.8V nominal for LiFePO4, but actual reading may vary.
8. **7. Wire PV to controller:** Connect PV leads to controller PV input with correct polarity. Keep panel covered/disconnected until ready to test.
9. **8. Meter PV side:** Before final PV connection or uncovering, verify PV open-circuit voltage and polarity. Call readings out loud.
10. **9. Wire inverter to battery side:** Wire inverter input to battery side through fuse/disconnect. Do not feed inverter from controller load terminals unless the equipment manual explicitly allows it.
11. **10. Peer check:** Neighbor or paired inspector traces full system and checks polarity, protection, crimp quality, and loose strands.
12. **11. Instructor approval:** Group explains circuit path and readings. Instructor approves before energizing.
13. **12. Commission:** Insert/close battery protection first, controller wakes up, uncover/connect PV, then turn inverter on. Test only with approved small load.
14. **13. Shutdown:** Turn off inverter, cover/disconnect PV, open/remove battery protection, then clean up and document results.

## Measurement log

| Check point | Expected / question | Measured value | Instructor initials |
|---|---|---|---|
| Battery direct | About 12.8V nominal LiFePO4; actual may vary | | |
| Controller BAT terminals | Same polarity as battery | | |
| PV panel open circuit | Polarity and voltage verified before connection | | |
| Controller PV input | Correct polarity | | |
| Inverter input | Battery-side protected DC input | | |
| AC output test load | Only with approved small load | | |

## Peer check before energizing

☐ Every + and - connection can be pointed to and named.

☐ Battery fuse/breaker is open or removed until final approval.

☐ PV panel is covered/face-down or disconnected until PV check.

☐ No loose strands, nicked conductors, or exposed copper outside terminals.

☐ Inverter input is battery-side through protection, not casually shoved into controller load terminals.

☐ Group can explain shutdown order.

## Shutdown order

- Turn inverter off and remove AC load.
- Cover or disconnect PV panel.

- Open/remove battery protection.
- Verify controller and inverter are no longer energized as needed.
- Clean kit and return all parts.

## Reflection

What mistake did your group catch before energizing?

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What measurement mattered most?

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What would you label better next time?

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